RESEARCH STATEMENT

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I am an algebraic geometer working in Hodge theory, especially nonabelian Hodge theory and its interactions with moduli spaces and topology of complex algebraic varieties. When $X \subseteq \mathbb{P}^n$ is a smooth complex algebraic variety, it's also a compact complex manifold with a natural Kähler metric. Methods of differential geometry then equip the singular cohomology groups with a Hodge decomposition

$$H^k(X, \mathbb{C}) = \bigoplus_{p+q=k} H^{p,q}(X)$$

where each $H^{p,q}(X)$ can be computed using algebraic differential forms. Therefore classical Hodge theory is a framework relating the topological, differential and algebraic perspectives. Nonabelian Hodge theory pushes this philosophy to cohomology with nonabelian coefficients, replacing $H^1(X, \mathbb{C})$ with $H^1(X, \mathrm{GL}_r) \coloneqq \mathrm{Hom}(\pi_1(X), \mathrm{GL}_r(\mathbb{C}))$, which is best understood as the moduli space of representations of the fundamental group $\pi_1(X)$. The theory's central correspondence uses harmonic theory to relate flat bundles, which encode topology, to Higgs bundles, which can be studied via algebraic methods like stability and are of independent interest. Subsequently, nonabelian Hodge theory has been very successful in resolving certain conjectures concerning fundamental group of an algebraic variety.

The picture is more complicated when X is not compact. As an example, the classical Hodge decomposition holds only on graded pieces of a filtration on $H^k(X, \mathbb{C})$. There has been substantial progress, in recent decades, on the nonabelian Hodge theory for quasiprojective varieties, but many results in the compact case are not yet extended. In the following section, my first two results are about extending known results to this new setting; the rest are applications of this theory to topological questions.

Results

1. **Nonabelian Hodge theory for smooth quasiprojective varieties.** For X a smooth projective variety, Simpson's correspondence [Sim92] identifies vector bundles with a holomorphic connection with semistable Higgs bundles with vanishing Chern classes. When restricted to polystable bundles, this is done by equipping the underlying C^∞-vector bundles with a harmonic metric, and gives a homeomorphism of the good moduli spaces $M_{\mathrm{DR}}(X, r)$ and $M_{\mathrm{Dol}}(X, r)$.

The homeomorphism transports a $\mathbb{C}^*$-action on $M_{\mathrm{Dol}}(X, r)$ to $M_{\mathrm{DR}}(X, r)$, which is complex analytically isomorphic to the character variety $M_{\mathrm{B}}(X, r) \coloneqq H^1(X, \mathrm{GL}_r)//\mathrm{GL}_r$; the quotient signifies forgetting the framing. This, in turn, equips rigid local systems (isolated points in $M_{\mathrm{B}}(X, r)$) with variation of Hodge structures. This imposes conditions on the monodromy groups, which Simpson used to rule out certain lattices such as $\mathrm{SL}_r(\mathbb{Z})$ as $\pi_1(X)$. Simpson's full correspondence was also crucial in Cao's result [Cao19] that the Albanese map is locally trivial.

Now suppose that $X = \bar{X} \backslash D$ is quasiprojective and smooth, with D a simple normal crossing divisor in smooth projective $\bar{X}$. A local system on X with unipotent local monodromy around D has a canonical extension to a logarithmic flat bundle on $\bar{X}$ whose residues are nilpotent. Its potential Higgs partner should be a logarithmic Higgs bundle with nilpotent residues and vanishing Chern classes. The boundary makes harmonic theory substantially harder, as singularity along D needs to be accounted for. Tame harmonic bundle theory, by Simpson [Sim90] and Mochizuki [Moc09], supplies the correspondence for polystable bundles. Two aspects of Simpson's compact theory were missing: a topological comparison of the corresponding moduli spaces, and a full extension-preserving correspondence.

Bakker-Brunebarbe-Tsimerman [BBT24] constructed a continuous bijection of the moduli spaces, and proved that it is a homeomorphism when X is a curve. I was able to prove that it is a homeomorphism in all dimensions.

Theorem A. For every smooth projective log pair $(\bar{X}, D)$, the nonabelian Hodge correspondence induces a homeomorphism

$$M^{\mathrm{nilp}}_{\mathrm{Dol}}(\bar{X}, D, r) \xrightarrow{\simeq} M^{\mathrm{nilp}}_{\mathrm{DR}}(\bar{X}, D, r)$$

between the good moduli space of polystable logarithmic Higgs bundles, with nilpotent residues and vanishing Chern classes, and the good moduli space of semisimple logarithmic flat bundles with nilpotent residues.

The proof is entirely algebraic; it reduces to the known curve case by proving that there is an abundance of sufficiently positive complete intersection curve $\bar{C} \subset \bar{X}$ where the restrictions give closed injective $M^{\mathrm{nilp}}_{\mathrm{Dol/DR}}(\bar{X}, D, r) \hookrightarrow M^{\mathrm{nilp}}_{\mathrm{Dol/DR}}(\bar{C}, D \cap C, r)$.

This moduli theorem still sees only polystable bundles. On a curve I obtain the following extension-preserving correspondence:

Theorem B. For a smooth projective log curve $(\bar{C}, D)$ there is an exact $\mathbb{C}-$linear equivalence of categories

$$\mathsf{Conn}_{\mathrm{nilp}}(\bar{C}, D) \xrightarrow{\simeq} \mathsf{Higgs}^{\mathrm{ss,deg}\,0}_{\mathrm{nilp}}(\bar{C}, D)$$

from all logarithmic flat bundles with nilpotent residues to all semistable degree 0 logarithmic Higgs bundles with nilpotent residues. It restricts to the tame nonabelian Hodge correspondence on polystable objects.

These extensions need not carry harmonic metrics. Proofs in the compact case [Sim92; Den21] crucially used the $\partial\bar{\partial}$-lemma which is not available on noncompact manifolds. My approach is moduli-theoretic, encoding extensions through deformation theory of the polystable bundles.

2. **Rank-two cohomological jump loci.** On a smooth projective variety X, the cohomological jump loci

$$\Sigma^i_k(X, r) := \left\{ [V] \in M_{\mathrm{B}}(X, r) \middle| \ \dim H^i(X, V) \geqslant k \right\}$$

measure where the cohomology of a rank r local system changes dimension. For the rank one case, $M_{\mathrm{B}}(X, 1)$ is a union of algebraic tori, and Green-Lazarsfeld [GL91] proved that irreducible components of Σ^i_k must be translates of subtori; Simpson [Sim93] improved this by showing that the translating points are torsion. Still in rank one, Budur-Wang [BW15] extended this to quasiprojective setting. These theorems give effective restrictions which rule out many finitely presented groups as fundamental groups of complex algebraic varieties [DPS09].

The first substantial difference in higher rank is that GL_r is not abelian and $M_{\mathrm{B}}(X, r)$ no longer carries the structure of an algebraic group. Goldman-Millson formality [GM88] describes the germ near a semisimple representation, and Hodge-theoretic refinements constrain the infinitesimal jump loci near a variation of Hodge structures. However, virtually nothing is known about the global geometry of $\Sigma^i_k(X, r)$ beyond rank $r = 1$. Rank-two case is the most promising due to a classification of SL_2-representations by Corlette-Simpson [CS08]. I was able to leverage their SL_2 result to study the GL_2-jump loci.

Theorem C. Every irreducible component $Z \subseteq \Sigma^i_k(X, 2)$ belongs to one of five monodromy types: reducible, irreducible dihedral, projectively finite, projectively dense rigid, or projectively dense nonrigid. The first four cases are described by rank-one jump loci and additionally extension cones or torsion twists. If Z is projectively dense nonrigid, its normalization Z^v is either an algebraic torus or a finite étale quotient

$$Z^v \simeq (Q^v \times T)/H$$

where Q is a fixed-determinant, fixed-inertia component of $M_{\mathrm{B}}(\mathcal{C}, 2)$ with $\mathcal{C}$ a proper hyperbolic Deligne-Mumford curve, and T an algebraic torus.

A key input to the theorem is a proposition which survives in higher rank: if $V \in \Sigma^i_k(X, r)$ underlies an admissible variation of mixed twistor structures, then $\widehat{\mathcal{O}}_{\Sigma^i_k(X,r),V}$ underlies a mixed twistor structure. This should, in principle, give additional restrictions on infinitesimal jump loci. I would also expect this rank-two result to rule out some fundamental groups missed by rank-one theory.

3. **Nonnegativity of Euler characteristic for Shimura varieties.** There is a famous open conjecture by Chern-Hopf-Thurston in topology, which predicts that for every closed aspherical $2n$-manifold X, $(-1)^n \chi(X) \geqslant 0$, i.e., the Euler characteristic has a definite sign. For a smooth complex n-manifold, the shifted constant sheaf $\mathbb{C}_X[n]$ is perverse, so nonnegativity of $\chi(X, \mathcal{P})$ for every perverse sheaf $\mathcal{P}$ is a natural strengthening.

Arapura-Wang [AW25] formulated and proved such positivity in several projective settings, the most general being the presence of a faithful rigid local system. For the quasiprojective case, Arapura-Patel [AP26] proved it for the moduli stacks $\mathcal{M}_{g,n}, \mathcal{A}_{g,n}$, and more generally Shimura varieties of Hodge type. They predicted that this should be true for all Shimura varieties, which I was able to confirm:

Theorem D. Let $X = \Gamma \backslash D$ be a finite volume quotient of a Hermitian symmetric domain by a torsion-free lattice. Then $\chi(X, \mathcal{P}) \geqslant 0$ for every complex perverse sheaf $\mathcal{P}$ on X. In particular, every Shimura variety has nonnegative Euler-Satake characteristic for every perverse sheaf.

The common feature of Arapura-Wang and Arapura-Patel is their use of the (logarithmic) Dubson-Kashiwara index theorem to reduce to proving nefness of certain cotangent bundles. In the projective setting, presence of a faithful rigid local system implies existence of a nice map into a period domain, whose negativity of sectional curvature gives the desired nefness. In the quasiprojective setting, they needed nefness of $T_{\bar{X}}(\log D)$ on some log smooth compactification $X \subset \bar{X}$. This follows from a result of Brunebarbe [Bru17] in the arithmetic case, and from Cadorel's [Cad21] in the nonarithmetic case.

Future Projects

1. **Log De Rham - Dolbeault cohomology comparison.** On a complex C^∞ vector bundle $\mathcal{E}$ on projective X with a flat C^∞-connection $\mathcal{D}$, if $\mathcal{E}$ has a harmonic metric then we can associate to it a flat connection and a Higgs bundle. There are then a De Rham complex and a Dolbeault complex, and Simpson [Sim92] proved that their hypercohomology groups are isomorphic. On quasiprojective $X = \bar{X} \backslash D$, we have similar logarithmic notions. There is a comparison result for those underlying admissible variations of mixed twistor structures, which were formerly the only way to pair a logarithmic flat connection with a logarithmic Higgs bundle. Now that there is a full correspondence for extensions, a natural question is whether the cohomology comparison holds for the new matchings.

Using the ideas in [KPT09], I have answered the question positively when $(\bar{X}, D)$ admits an iterated fibration of smooth projective log curves. If we are not in the log setting, this is usually enough to get the general case, as any smooth quasiprojective variety can be covered by tower of affine curves. Such a cover is not available here, but it might suggest that we can modify $(\bar{X}, D)$ in a controlled way to get such a fibration.

2. **Parabolic nonabelian Hodge correspondence.** Unipotent monodromy is a natural first case, but even if one cares only about local systems of geometric origin, these generally have quasi-unipotent monodromy around the boundary D. Their correct Dolbeault counterparts are parabolic Higgs bundles with rational weights and vanishing Chern classes. Mochizuki's tame harmonic correspondence does treat this more general class of polystable parabolic bundles.

I plan to extend both results in Section 1 to this parabolic setting. Given any local system with quasi-unipotent monodromy around D, a finite cover of $\bar{X}$ turns the local system back to unipotent local monodromy. For moduli applications, we need a systematic and local-system-independent way of doing these covers; the natural choice is a root stack. This idea of turning a parabolic bundle into a vector bundle on a root stack has seen successes in [Bor07; BV12]. To carry out this approach, I will develop the necessary harmonic and twistor theory on such a stack, which seems promising considering [Sim11].

3. **Rank-three cohomological jump loci.** After a structure theorem for rank-two jump loci, rank-three is the natural next step. This case is substantially harder, as there is no classification theorem for nonrigid representations. However, as part of Section 2, I proved that jump loci in all ranks are Hodge substack in the sense of [BBT24]. As a consequence, for an irreducible component $Z \subseteq \Sigma^i_k(X, 3)$, there is a Shafarevich map

$$\mathrm{sh}_Z : X \to \mathcal{Y}_Z$$

to a Deligne-Mumford stack, such that every local system $[V] \in Z$ is pulled back from a local system on $\mathcal{Y}_Z$, i.e., $V = \mathrm{sh}_Z^* W$. It follows that

$$\mathrm{R}\Gamma(X, V) = \mathrm{R}\Gamma(\mathcal{Y}_Z, W \otimes \mathrm{R}(\mathrm{sh}_Z)_* \mathbb{C}_X)$$

and we can instead study the twisted jump loci on $\mathcal{Y}_Z$. One would hope that by imposing some conditions on X, $\mathcal{Y}_Z$ would be simple enough for our purpose. Or more optimistically, for rank-three, we can bound the dimension of $\mathcal{Y}_Z$ when Z is projectively positive-dimensional. An ingredient of Section 2 is an analysis of these twisted jump loci on a DM curve; a related question here would be a similar analysis for DM surfaces.

4. **Comparing Hodge-theoretic and KSBA moduli compactifications.** Hodge-theoretic moduli spaces show up in the presence of a Torelli theorem; informally, this happens when varieties can be distinguished by their Hodge structures. Examples include abelian varieties, K3 surfaces, and generically hypersurfaces in $\mathbb{P}^n$. In these cases, we can take the moduli space to be the image of a period map. Classically we have the Baily-Borel-Satake [BB66] compactification when the period domain is Hermitian symmetric (abelian varieties and K3 surfaces included). Recently, a Baily-Borel compactification in much more generality was constructed in [Bak+25].

The main feature of these compactifications is that their boundaries record degeneration of Hodge structures. This is not a modular compactification, as it does not give the limit as an algebraic variety. Thus an interesting question is comparing these Hodge-theoretic compactifications with modular compactifications, a prominent example of which is the KSBA moduli space. There are compelling precedents. For principally polarized abelian varieties, the main component of Alexeev's compactification by semiabelic pairs [Ale02] is a toroidal compactification of the period image. For polarized K3 surfaces, Alexeev and Engel [AE23] proved that certain KSBA compactification is semitoroidal. In both cases, the KSBA compactification retains geometric data that the Baily-Borel compactification contracts; it can be thought of as a blow-up.

I plan to study this question for canonically polarized hypersurfaces, i.e., $X_d \coloneqq V(P) \subseteq \mathbb{P}^{n+1}$ with $\deg P \geqslant n+3$. [Bak+25] provides a compactification for moduli of these hypersurfaces. The starting point would be understanding one-parameter degenerations; the first interesting case might simply be quintic surfaces.

References

[AE23] Valery Alexeev and Philip Engel. "Compact moduli of K3 surfaces". In: *Annals of Mathematics* 198.2 (2023), pp. 727–789.

[Ale02] Valery Alexeev. "Complete moduli in the presence of semiabelian group action". In: *Annals of mathematics* (2002), pp. 611–708.

[AP26] Donu Arapura and Deepam Patel. "Nonnegativity of signed Euler characteristics of moduli of curves and abelian varieties: D. Arapura, D. Patel". In: *Mathematische Annalen* 394.2 (2026), p. 34.

[AW25] Donu Arapura and Botong Wang. "Perverse sheaves on varieties with large fundamental groups". In: *Journal of Differential Geometry* 129.1 (2025), pp. 1–15.

[Bak+25] Benjamin Bakker et al. *Baily–Borel compactifications of period images and the b-semiampleness conjecture*. 2025. arXiv: `2508.19215 [math.AG]`.

[BB66] Walter L Baily and Armand Borel. "Compactification of arithmetic quotients of bounded symmetric domains". In: *Annals of mathematics* 84.3 (1966), pp. 442–528.

[BBT24] Benjamin Bakker, Yohan Brunebarbe, and Jacob Tsimerman. *The linear Shafarevich conjecture for quasiprojective varieties and algebraicity of Shafarevich morphisms*. 2024. arXiv: `2408.16441 [math.AG]`.

[Bor07] Niels Borne. "Fibrés Paraboliques et Champ des Racines". In: *International Mathematics Research Notices* 2007 (Jan. 2007), rnm049. ISSN: 1073-7928.

[Bru17] Yohan Brunebarbe. "Semi-positivity from Higgs bundles". In: *arXiv preprint arXiv:1707.08495* (2017).

[BV12] Niels Borne and Angelo Vistoli. “Parabolic sheaves on logarithmic schemes”. In: *Advances in Mathematics* 231.3-4 (2012), pp. 1327–1363.

[BW15] Nero Budur and Botong Wang. “Cohomology jump loci of quasi-projective varieties”. In: *Annales scientifiques de l’École Normale Supérieure* Ser. 4, 48.1 (2015), pp. 227–236.

[Cad21] Benoit Cadorel. “Symmetric differentials on complex hyperbolic manifolds with cusps”. In: *Journal of Differential Geometry* 118.3 (2021), pp. 373–398.

[Cao19] Junyan Cao. “Albanese maps of projective manifolds with nef anticanonical bundles”. In: *Annales Scientifiques de l’École Normale Supérieure*. Vol. 52. 5. 2019, pp. 1137–1154.

[CS08] Kevin Corlette and Carlos Simpson. “On the classification of rank-two representations of quasiprojective fundamental groups”. In: *Compositio Mathematica* 144.5 (2008), pp. 1271–1331.

[Den21] Ya Deng. “A note on the Simpson correspondence for semistable Higgs bundles”. In: *Pure Appl. Math. Q.* 17.5 (2021), pp. 1899–1911.

[DPS09] Alexandru Dimca, Ştefan Papadima, and Alexander I. Suciu. “Topology and geometry of cohomology jump loci”. In: *Duke Mathematical Journal* 148.3 (2009), pp. 405–457.

[GL91] Mark Green and Robert Lazarsfeld. “Higher obstructions to deforming cohomology groups of line bundles”. In: *Journal of the American Mathematical Society* 4.1 (1991), pp. 87–103.

[GM88] William M Goldman and John J Millson. “The deformation theory of representations of fundamental groups of compact Kähler manifolds”. In: *Publications Mathématiques de l’IHÉS* 67 (1988), pp. 43–96.

[KPT09] Ludmil Katzarkov, Tony Pantev, and Bertrand Toën. “Algebraic and topological aspects of the schematization functor”. In: *Compositio mathematica* 145.3 (2009), pp. 633–686.

[Moc09] Takuro Mochizuki. “Kobayashi–Hitchin correspondence for tame harmonic bundles II”. In: *Geometry & Topology* 13.1 (2009), pp. 359–455.

[Sim11] Carlos Simpson. “Local systems on proper algebraic V-manifolds”. In: *Pure and Applied Mathematics Quarterly* 7.4 (2011), pp. 1675–1760.

[Sim90] Carlos T Simpson. “Harmonic bundles on noncompact curves”. In: *Journal of the American Mathematical Society* 3.3 (1990), pp. 713–770.

[Sim92] Carlos T Simpson. “Higgs bundles and local systems”. In: *Publications Mathématiques de l’IHÉS* 75 (1992), pp. 5–95.

[Sim93] Carlos Simpson. “Subspaces of moduli spaces of rank one local systems”. In: *Annales scientifiques de l’Ecole normale supérieure*. Vol. 26. 3. 1993, pp. 361–401.